MILITARY STANDARD

TRUCKLOADING
TORPEDO MK 48 MOD 1
IN CONTAINER MK 481 MODS 0 & 1

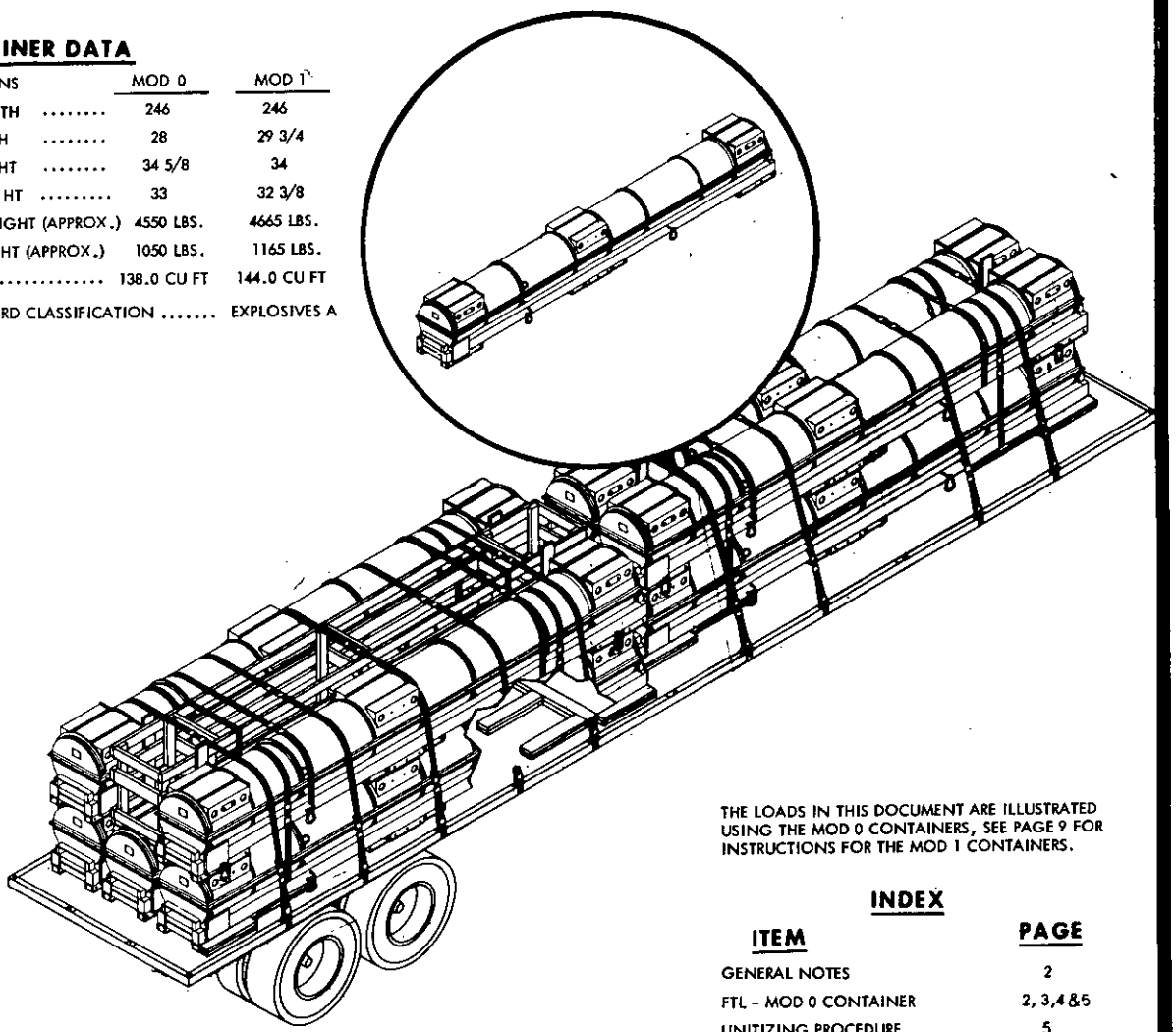
MIL-STD-1320-113C (NAVY)

25 SEPTEMBER 1985

SUPERSEDING
MIL-STD-1320-113B (NAVY)
15 APRIL 1980

CONTAINER DATA

DIMENSIONS	MOD 0	MOD 1
LENGTH	246	246
WIDTH	28	29 3/4
HEIGHT	34 5/8	34
STACKING HT	33	32 3/8
GROSS WEIGHT (APPROX.)	4550 LBS.	4665 LBS.
TARE WEIGHT (APPROX.)	1050 LBS.	1165 LBS.
CUBE	138.0 CU FT	144.0 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVES A	



THE LOADS IN THIS DOCUMENT ARE ILLUSTRATED USING THE MOD 0 CONTAINERS, SEE PAGE 9 FOR INSTRUCTIONS FOR THE MOD 1 CONTAINERS.

INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL - MOD 0 CONTAINER	2, 3, 4 & 5
UNITIZING PROCEDURE	5
FTL - MOD 0 CONTAINER	6 & 7
LTL - MOD 0 CONTAINER	8
FTL & LTL - MOD 1 CONTAINER	9

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINER LOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

REV LTR	REVISION DESCRIPTION	DATE	APPROVAL: NDA	APPROVAL: SYSCOM
C	ADDED ONE HIGH LOAD FOR 4-6 CONTAINERS	25 SEP 85		
B	ADDED FTL & LTL'S FOR MOD 1 CONTAINER	15 APR 80		
A	ADDED SEPARATOR ASSY, DETAIL H	30 NOV 78		

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENT ONLY

M. S. Gray NWHC 4/9/80
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

A. E. Kelly 4/9/80
SIGNATURE SEASYCOM / BY DIRECTION DATE

DISTRIBUTION STATEMENT A
Approved for public release; distribution is unlimited.

ORIGINATOR L. R. Quingi 4/9/80
DATE

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 2026-10-23 23:44Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

1. THE DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE MK 48 MOD 1 TORPEDO WHEN THE TORPEDO IS IN THE MK 481 MOD 0 AND 1 CONTAINER.
2. PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44, 45 and 48 FT. FLATBED TRAILERS. DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS.
3. STACKED CONTAINERS MUST BE UNITIZED. FOR UNITIZING INSTRUCTIONS REFER TO PAGE 5. WHEN A STACK OF CONTAINERS IS TWO HIGH, THE TOP LAYER SHALL BE SECURED TOGETHER WITH TWO 1 1/4 X .035 CROSS STRAPS. THESE STRAPS ENCIRCLE THE TOP CONTAINERS SECURING THEM TOGETHER.

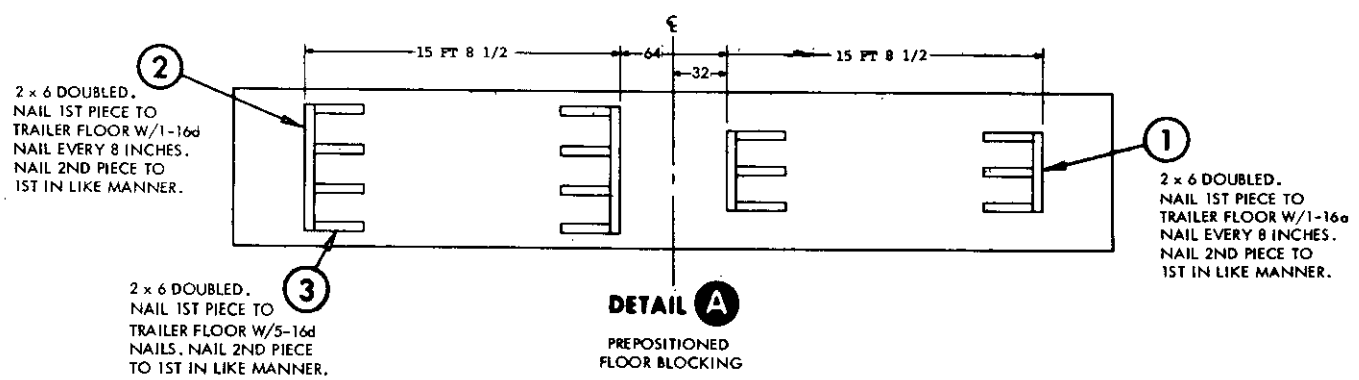
WARNING

TO AVOID TOPPLING, SPECIAL CARE SHOULD BE TAKEN WHEN MOVING UNITIZED CONTAINERS. THE UNITIZING OF THE CONTAINERS IS AUTHORIZED AS A PART OF THE TRUCKLOADING PROCEDURE ONLY. UNITIZING IS TO ENSURE CONTINUOUS ENGAGEMENT OF THE STACKING FEATURES AND MOVEMENT OF THE UNITIZED CONTAINERS SHOULD BE LIMITED TO THAT NECESSARY TO LOAD OR UNLOAD THE TRAILER. CONTAINERS SHALL BE DEUNITIZED AFTER UNLOADING THE VEHICLE.

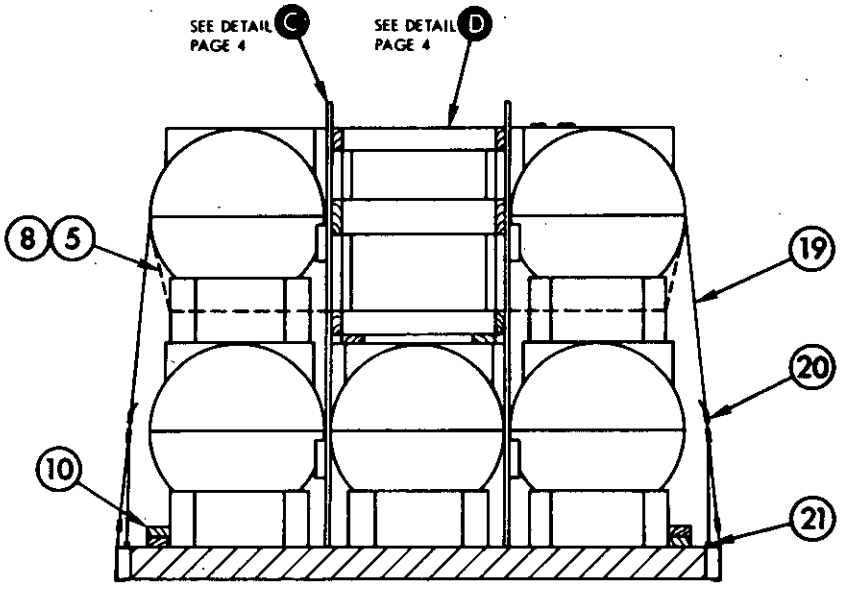
4. IF SUITABLE CAPACITY HANDLING EQUIPMENT IS NOT AVAILABLE FOR MOVING UNITIZED TWO HIGH CONTAINERS (TOTAL WEIGHT 9100 LBS). IT WILL BE NECESSARY TO UNITIZE THE CONTAINERS IN PLACE.
5. CHAINS AND STRAPS MAY BE USED INTERCHANGEABLY FOR TIE-DOWNS ON A 1 TO 1 BASIS. WHEN USING CHAIN, LADING SHALL BE PROTECTED FROM DAMAGE BY THE CHAIN LINKS.
6. STEEL STRAP USED AS TIEDOWNS SHALL BE 2 X .050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE BASIC TRUCKLOADING DOCUMENT MIL-STD-1320 AND AS SHOWN ON PAGE 3 OF THIS DOCUMENT.
7. CHAINS, FITTINGS AND LOADBINDERS SHALL MEET THE REQUIREMENTS OF THE BASIC TRUCKLOADING DOCUMENT MIL-STD-1320 AND AS SHOWN ON PAGE 6 OF THIS DOCUMENT FOR BOTH ONE HIGH AND TWO HIGH CONTAINERS.
8. EACH STACK OF CONTAINERS SHALL HAVE A MINIMUM OF TWO TIEDOWNS.
9. PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY NAVSEA OP 2165 AND REPORTED ON DD FORM 626. THE CHAINS, FITTINGS, AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.
10. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS, AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA AND ATTACH "EXPLOSIVES A" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
12. UNLESS OTHERWISE SPECIFIED ALL MATERIAL SHALL BE AS SPECIFIED IN THE GENERAL DOCUMENT MIL-STD-1320.
13. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MILITARY STANDARD 1320.

FTL 44/45/48 FT FLATBED TRAILERS (MK 481 MOD 0 CONTAINERS)

1. 44, 45, AND 48 FT FLATBED TRAILERS ARE CONSIDERED SPECIAL EQUIPMENT AND ARE NOT ALWAYS GENERALLY AVAILABLE.
2. WHEN 44, 45, 48 FT TRAILERS ARE NOT AVAILABLE, SHIPMENT SHALL BE MADE ON 40 OR 42 FT. TRAILERS. (SEE PAGES 7 & 8).
3. FILLER ASSEMBLY, DETAILED, SHALL BE STRAPPED TO LOWER CONTAINER SIMILAR TO UNITIZING PROCEDURE SHOWN ON PAGE 4.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.



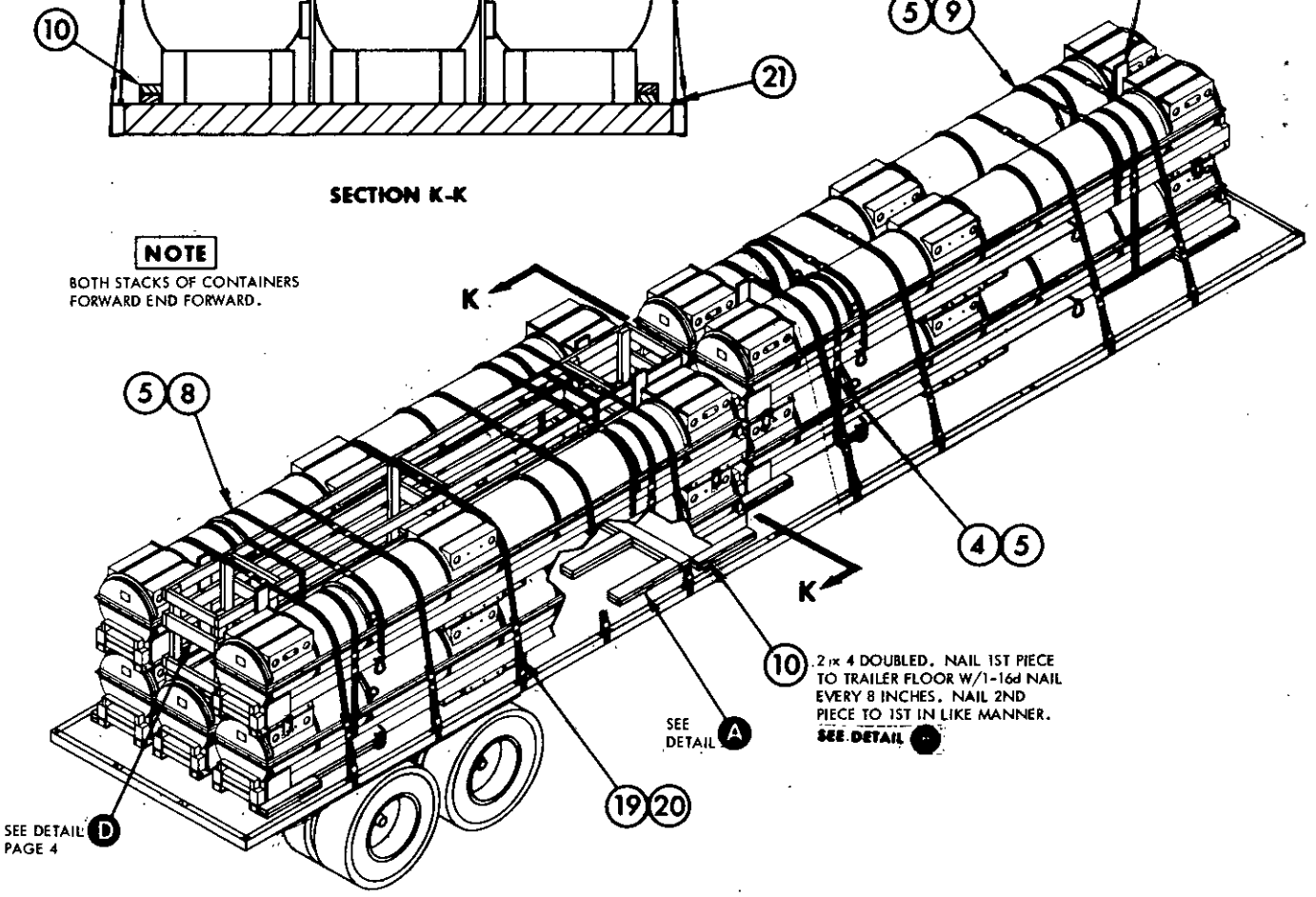
SECTION K-K

TRUCKLOAD DATA

NUMBER OF CONTAINERS 9
WEIGHT (APPROX.) - MOD 0 ..... 40,950 LBS.
WEIGHT (APPROX.) - MOD 1 ..... 41,985 LBS.

NOTE

BOTH STACKS OF CONTAINERS FORWARD END FORWARD.



10 2 x 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER. SEE DETAIL B

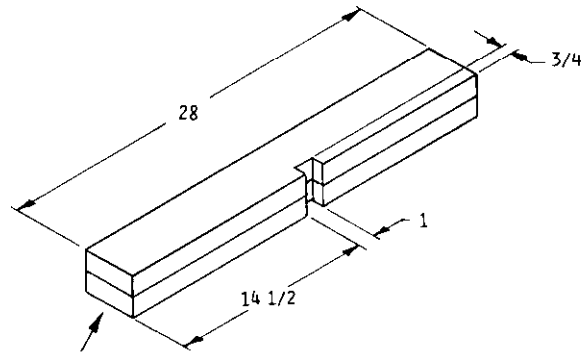
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
21	STAKE POCKET PAD (STRAPPING)	2 x .050 x 18	18	-	-	-
20	SEAL	FOR 2" STRAP	72	-	-	-
19	TIE-DOWN STRAP	2 x .050 x 28 FT	9	-	-	-
18	SUPPORT BRACE	2 x 4 x 24 3/4	2	SEE 14	-	-
17	SUPPORT	2 x 4 x 23	4	12, 18	2/JOINT	16d
16	BUFFER PIECE	1 x 6 x 14 FT	2	15	1/FT	10d
15	LONGITUDINAL	2 x 6 x 19 FT 4	2	13	1/JOINT	16d
14	LONGITUDINAL	2 x 4 x 19 FT 4	4	18 12	2/JOINT 1/JOINT	16d 16d
13	CROSS BRACE	2 x 6 x 24 3/4	5	11	3/JOINT	16d
12	HORIZONTAL	2 x 4 x 24 3/4	10	11	2/JOINT	16d

LIST OF MATERIALS AND NAILING DATA

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
11	VERTICAL	2 x 4 x 33	10	14, 15	2/JOINT	10d
10	SLEEPER	2 x 4 x 28	16	SEE DETAIL B		
9	CROSS STRAP	1 1/4 x .035 x 18 FT	2	-	-	-
8	CROSS STRAP	1 1/4 x .035 x 23 FT	2	-	-	-
7	CLEAT	2 x 2 x 6	24	SEE 6	-	-
6	UPRIGHT	1 x 6 x 72	6	7	3/JOINT	6d
5	SEAL	FOR 1 1/4 STRAP	28	-	-	-
4	UNITIZING STRAP	1 1/4 x .035 x 18 FT	10	-	-	-
3	BACKUP CLEAT	2 x 6 x 30	28	SEE DETAIL A		
2	CROSS MEMBER	2 x 6 x 79	4	SEE DETAIL A		
1	CROSS MEMBER	2 x 6 x 50	4	SEE DETAIL A		

LIST OF MATERIALS AND NAILING DATA

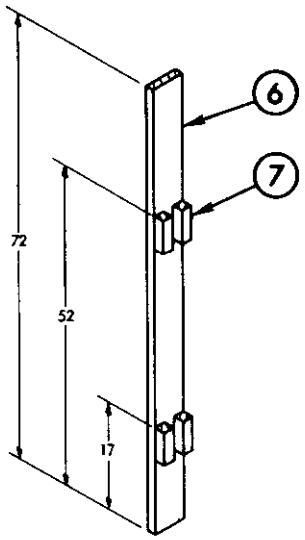
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.



DETAIL B

DOUBLE, NOTCHED SLEEPER
2 X 4 X 28

NOTE: THIS SLEEPER ASSEMBLY IS REQUIRED FOR MOD 0 CONTAINERS AND ALSO FITS MOD 1 CONTAINERS. SHIPMENTS OF MOD 1 CONTAINERS AS BOTTOM LAYER DO NOT REQUIRE NOTCHED SLEEPERS WHILE OTHER DIMENSIONS ARE UNCHANGED

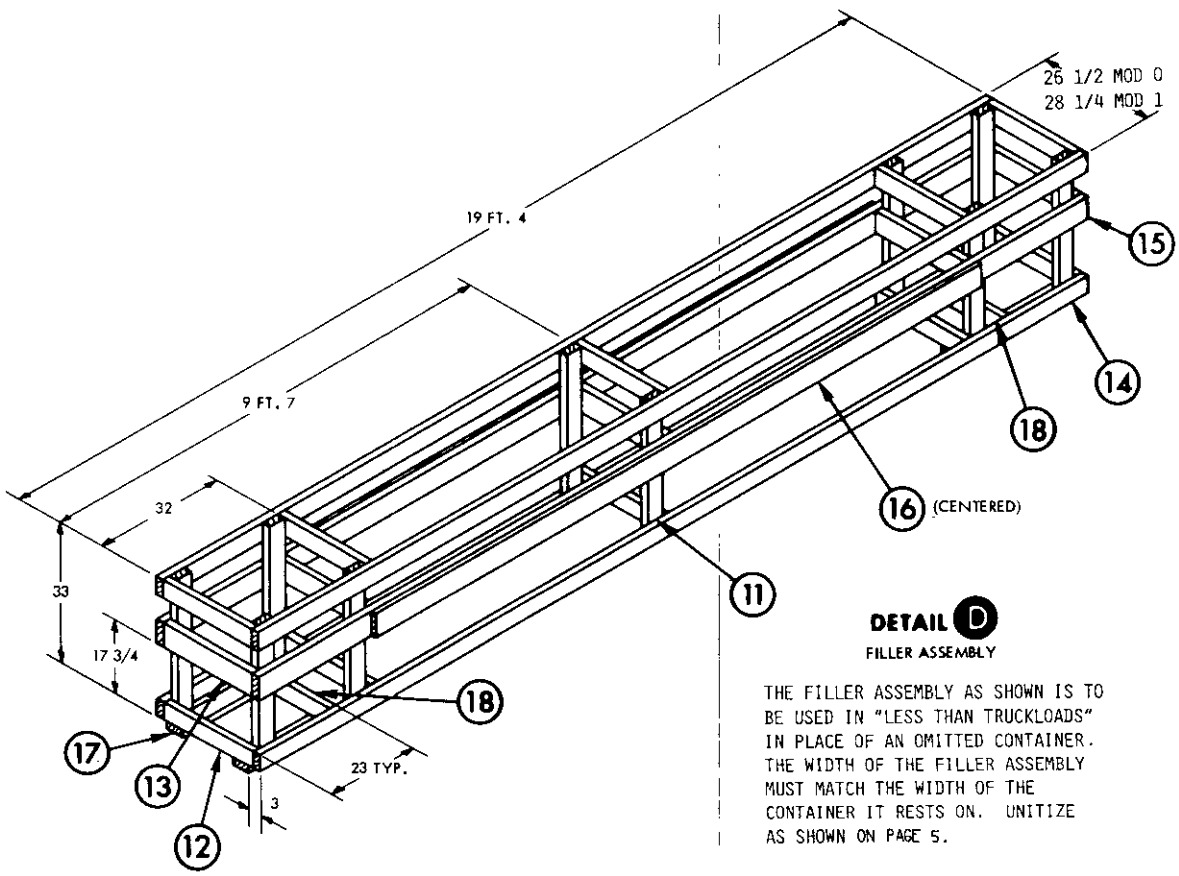


DETAIL C

ANTI-CHAFING FRAME

6 REQUIRED

THE ANTI-CHAFING FRAME MUST BE USED WITH ALL LOADS OF MOD 0 AND WITH MIXED LOADS OF MOD 0 AND MOD 1 CONTAINERS. THEY ARE NOT REQUIRED FOR LOADS OF MOD 1 CONTAINERS ONLY.



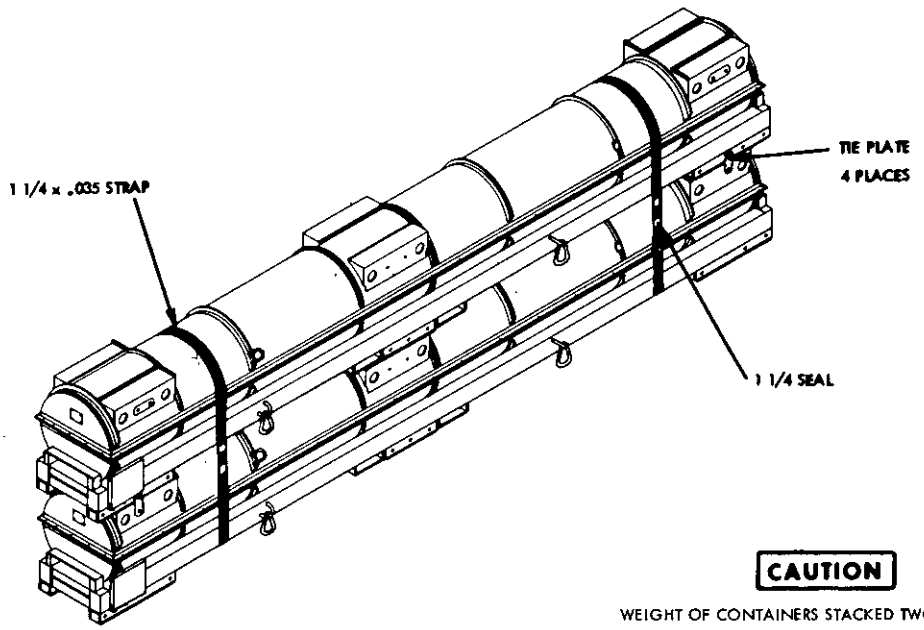
DETAIL D

FILLER ASSEMBLY

THE FILLER ASSEMBLY AS SHOWN IS TO BE USED IN "LESS THAN TRUCKLOADS" IN PLACE OF AN OMITTED CONTAINER. THE WIDTH OF THE FILLER ASSEMBLY MUST MATCH THE WIDTH OF THE CONTAINER IT RESTS ON. UNITIZE AS SHOWN ON PAGE 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.

UNITIZING PROCEDURE



CAUTION

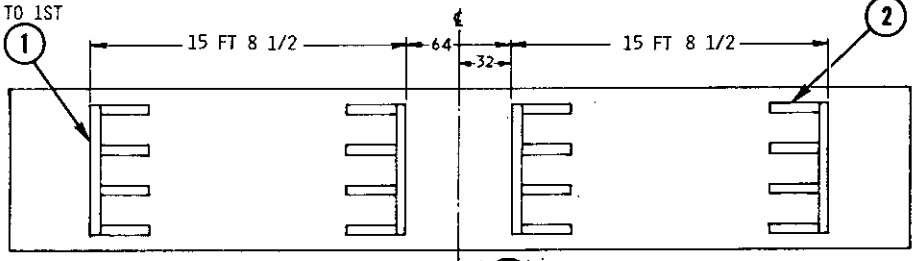
WEIGHT OF CONTAINERS STACKED TWO HIGH IS 9100 LBS. IF ADEQUATE CAPACITY HANDLING EQUIPMENT IS NOT AVAILABLE, CONTAINERS MUST BE UNITIZED IN PLACE ON THE TRAILER.

CONTAINER UNITIZING PROCEDURE

1. STACK TWO CONTAINERS USING A FORK LIFT TRUCK.
2. BE CERTAIN THAT THE NESTING FEATURES AND BEARING SURFACES ARE PROPERLY POSITIONED TO PREVENT SHIFTING IN LATERAL AND LONGITUDINAL DIRECTIONS.
3. FASTEN TIE PLATES IN POSITION ON CONTAINERS. (MOD 0 CONTAINERS ONLY).
4. SECURE BOTTOM CONTAINER TO TOP CONTAINER WITH TWO 1 1/4 X .035 STEEL STRAPS POSITIONED AS INDICATED ABOVE.
5. AFTER TENSIONING STRAPS, SEAL WITH ONE DOUBLE NOTCHED SEAL.
6. CONTAINERS SHALL BE DEUNITIZED AFTER UNLOADING THE VEHICLE.

2 X 6 DOUBLED
NAIL FIRST PIECE TO
TRAILER FLOOR W/1-16d
NAIL EVERY 8 INCHES
NAIL 2ND PIECE TO 1ST
IN LIKE MANNER. (1)

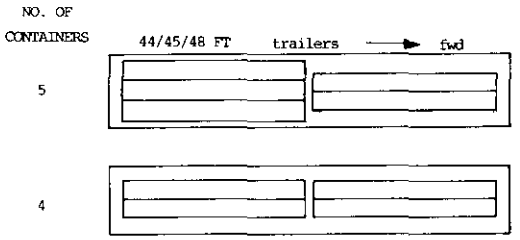
2 X 6 DOUBLED.
NAIL 1ST PIECE TO
TRAILER FLOOR w/5-16d
NAILS. NAIL 2ND PIECE
TO 1ST IN LIKE MANNER. (2)



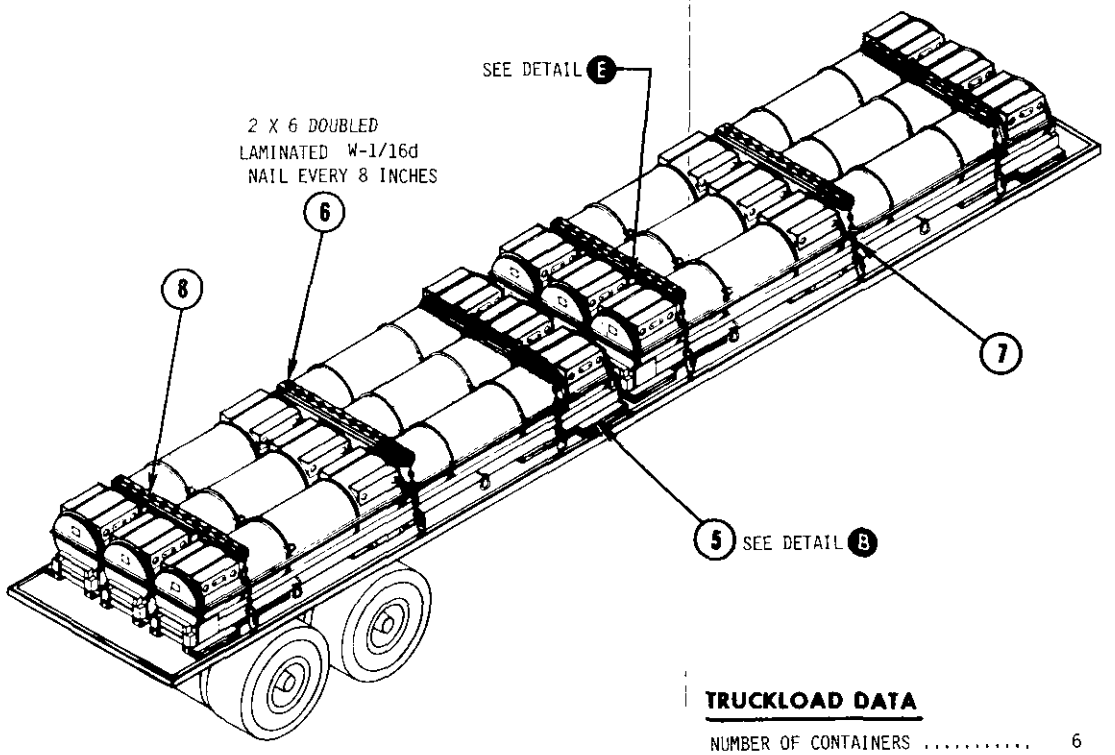
DETAIL G

PREPOSITIONED FLOOR BLOCKING
FOR ONE CONTAINER HIGH, LOAD OF 6

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.

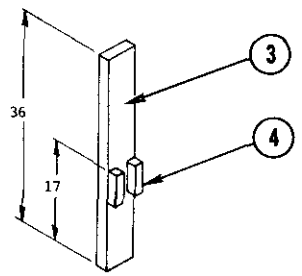


PREFERRED LTL FOR 6, 5, OR 4 CONTAINERS
REQUIRES 44/45 /48 FOOT TRAILER



TRUCKLOAD DATA

NUMBER OF CONTAINERS 6
WEIGHT (MOD 0) .....27,300 LBS
WEIGHT (MOD 1) ..... 27,990 LBS



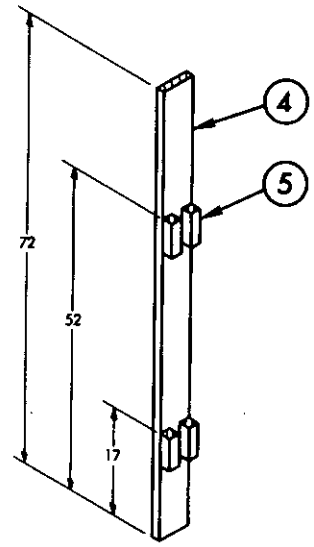
ANTI CHAFTING FRAME
8 REQUIRED

DETAIL E

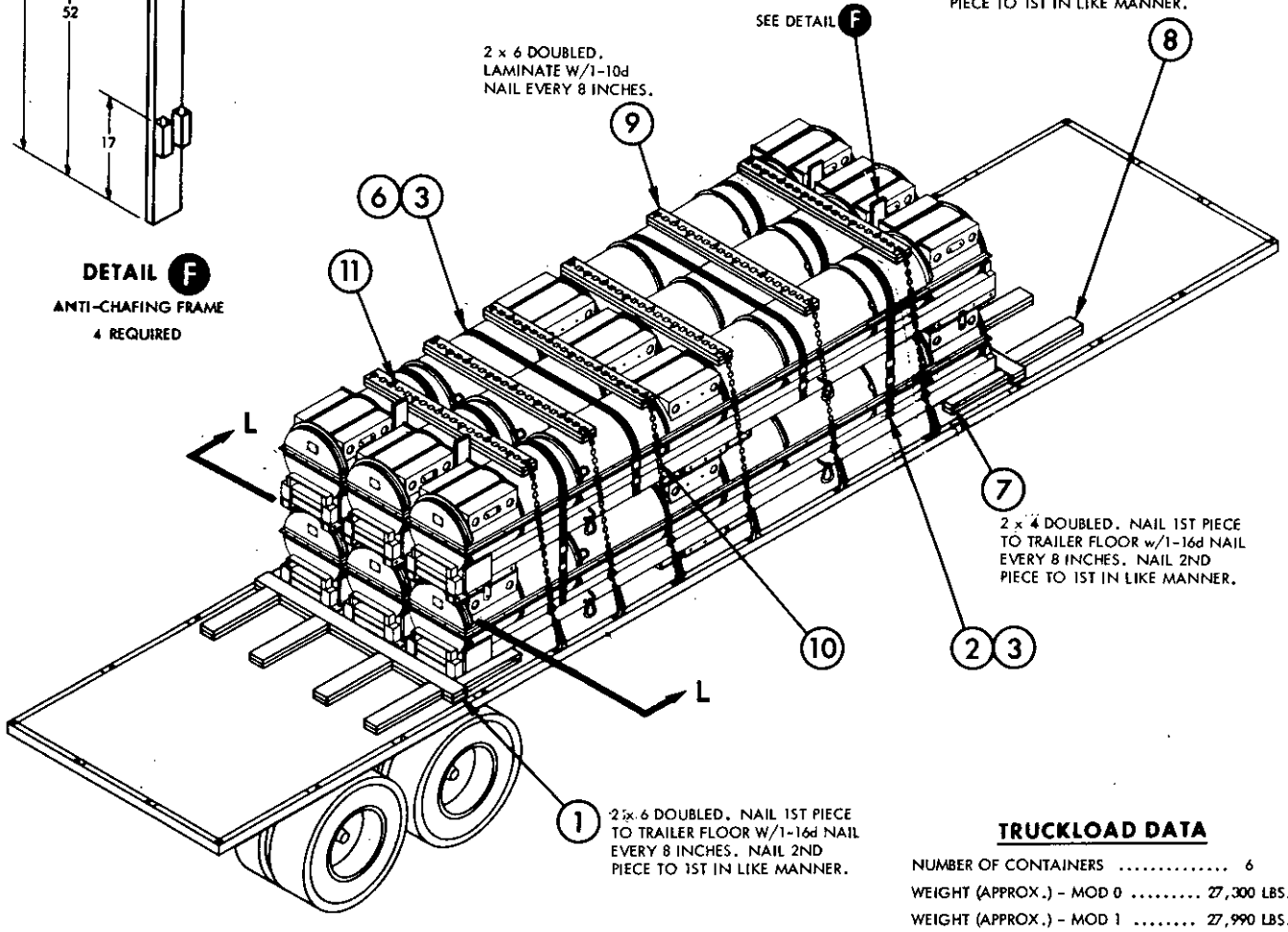
ITEM NO.	DESCRIPTION	DIMENSIONS	NO. REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
8	NAILS	20d	24	6	THROUGH CHAIN LINK AND CLINCHED	
7	CHAIN AND LOAD BINDERS	5/16 OR 3/8	6	SEE GENERAL NOTES		
6	PROTECTOR BOARD	2 X 6 X 73	12	SEE FIELD NOTE		
5	SLEEPER	2 X 4 X 28	16	SEE DETAIL B		
4	CLEAT	2 X 2 X 6	16	SEE 4		
3	UPRIGHT	1 X 6 X 36	8	5	3/JOINT	6d
2	BACKUP CLEAT	2 X 6 X 30	32	SEE FIELD NOTE		
1	CROSSMEMBER	2 X 6 X 79	8	SEE FIELD NOTE		

LIST OF MATERIALS

Source: https://assist.dla.mil -- Downloaded: 2026-10-23 12:34:42
Check the source to verify that this is the current version before use.



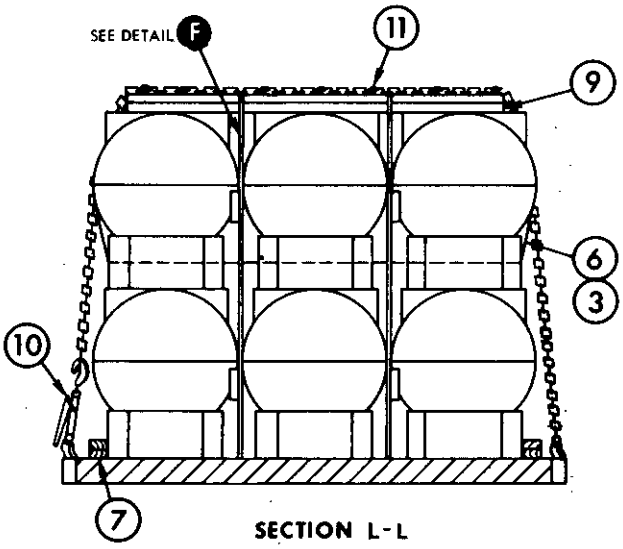
DETAIL F
ANTI-CHAFING FRAME
4 REQUIRED



TRUCKLOAD DATA

NUMBER OF CONTAINERS 6
WEIGHT (APPROX.) - MOD 0 ..... 27,300 LBS.
WEIGHT (APPROX.) - MOD 1 ..... 27,990 LBS.

NOTE: USE THIS LOAD AND THE LTL'S FOR 4 AND 5 CONTAINERS ONLY IF 44/45/48 FOOT TRAILERS ARE NOT AVAILABLE.



SECTION L-L

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
11	NAILS	20d	24	9	THROUGH CHAIN LINK & CLINCHED	
10	CHAIN & LOAD BINDER	5/16 OR 3/8	6	SEE GENERAL NOTES		
9	PROTECTOR BOARD	2 x 6 x 73	12	SEE FIELD NOTE		
8	END CLEAT	2 x 6 x 30	16	SEE FIELD NOTE		
7	SLEEPER	2 x 4 x 30	8	SEE FIELD NOTE		
6	CROSS STRAP	1 1/4 x .035 x 23 FT	2	-	-	-
5	CLEAT	2 x 2 x 6	16	SEE 4	-	-
4	UPRIGHT	1 x 6 x 72	4	5	3/JOINT	6d
3	SEAL	FOR 1 1/4 STRAP	16	-	-	-
2	UNITIZING STRAP	1 1/4 x .035 x 18 FT	6	-	-	-
1	CROSS MEMBER	2 x 6 x 90	4	SEE FIELD NOTE		

LIST OF MATERIALS AND NAILING DATA

Source: https://assist.dla.mil - Downloaded: 2020-12-21T14:47Z
Check the source to verify that this is the current version before use.

LTL 44/45/48 FT AND 42/40 FT FLATBED TRAILERS (MK 481 MOD 0 CONTAINERS)

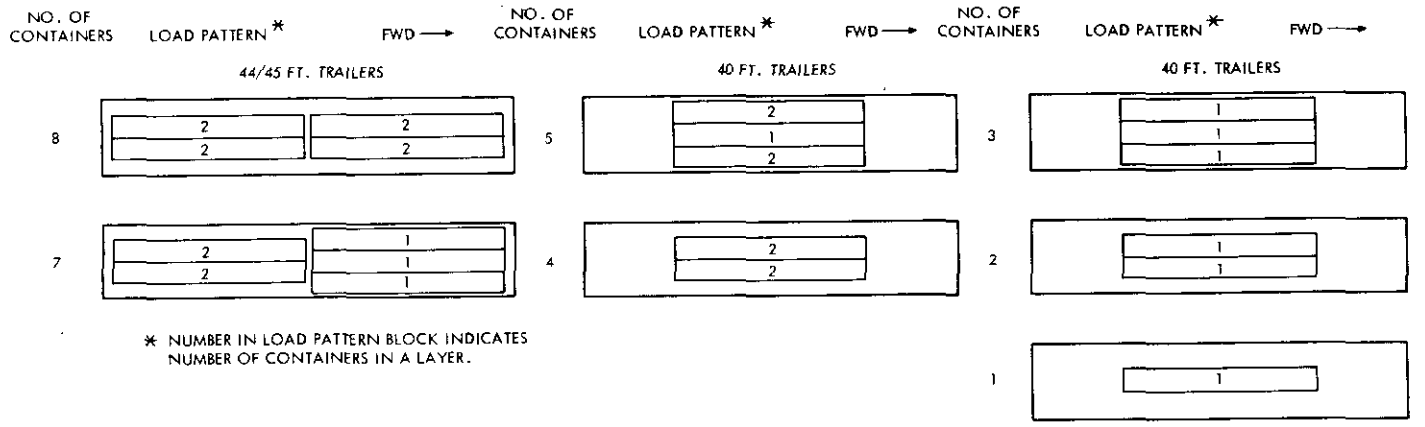
1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, SELECT THE CORRECT-LOAD PATTERN FROM TABLE 1 FOR THE NUMBER OF ITEMS TO BE SHIPPED.

NOTE

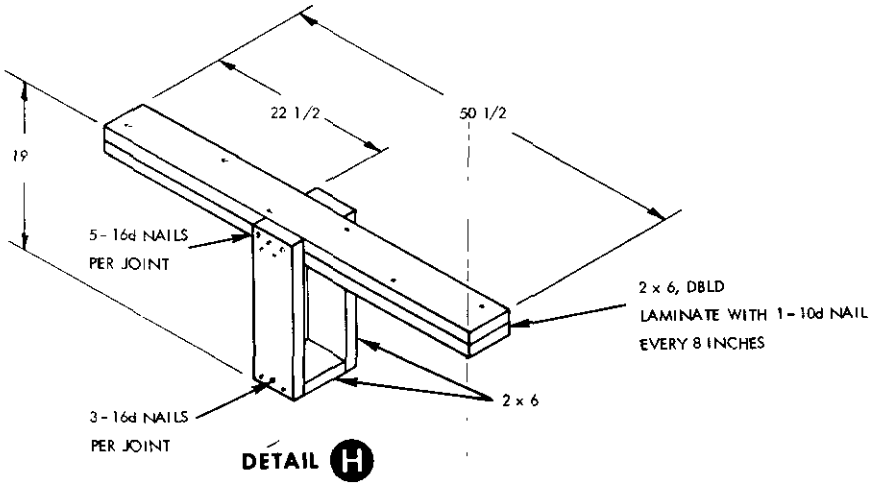
WHEN SHIPPING 7 OR 8 CONTAINERS, A 44/45 FT. FLATBED TRAILER IS REQUIRED.
WHEN SHIPPING 4, 5, OR 6 CONTAINERS A SINGLE LAYER ON A 44/45/48 FOOT TRAILER IS PREFERRED.
IF THE LONGER TRAILERS ARE NOT AVAILABLE, THEN USE THE LTL'S SHOWN BELOW AND THE FTL SHOWN ON PAGE 7.
WHEN SHIPPING 1 TO 3 CONTAINERS A 42/40 FT. FLATBED TRAILER SHOULD BE USED.

2. BLOCK, BRACE AND TIE-DOWN LTL LOADS USING THE APPLICABLE PRINCIPLES SHOWN ON PAGES 3 AND 5. SEE GENERAL NOTE FOR QUANTITY OF THE TIE-DOWNS TO BE USED.

TABLE I



3. SHIPPING REQUIREMENTS MAY ARISE WHEN SHIPPING TWO CONTAINERS TO SEPARATE THE CONTAINERS BY A SUFFICIENT AMOUNT TO PERMIT ACCESS TO THE CONTAINER LATCHES. THE TWO CONTAINERS SHOULD BE LOADED AS SHOWN IN TABLE I USING THE PROTECTOR BOARD/SEPARATOR, DETAIL H, BETWEEN CONTAINERS. LOCATE THE SEPARATORS APPROXIMATELY 28 INCHES FROM EACH END OF THE STACK WITH THE DOUBLED 2 x 6 LAYING ACROSS THE TOPS OF THE CONTAINERS. POSITION TIEDOWNS OVER THE DOUBLED 2 x 6.



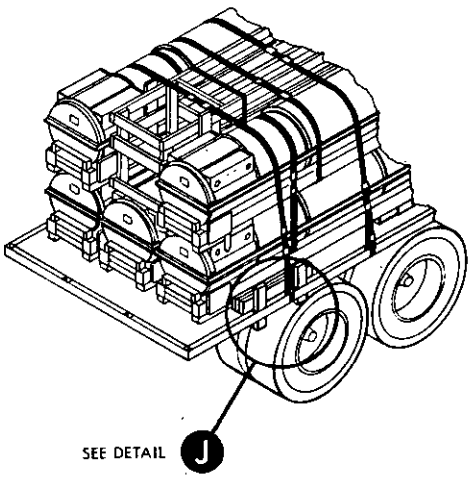
DETAIL H

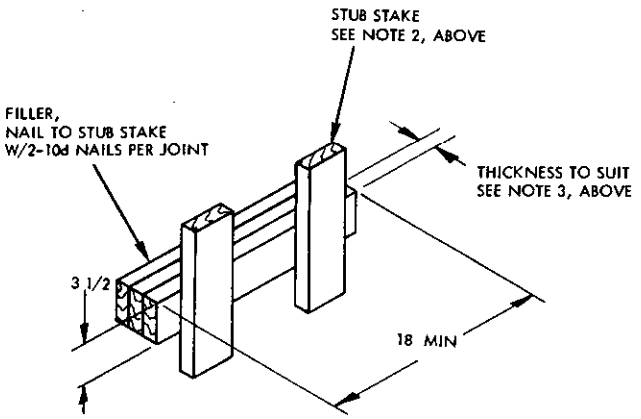
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.

FTL & LTL 44/45/48 FT & 40/42 FT FLATBED TRAILERS (MK 481 MOD 1 CONTAINERS)

INSTRUCTIONS FOR LOADING MK 481 MOD 1 CONTAINERS

1. THE MK 481 MOD 1 CONTAINERS SHALL BE LOADED USING THE SAME PROCEDURES AS FOR THE MK 481 MOD 0 WITH THE FOLLOWING EXCEPTIONS:
   a. THE ANTI-CHAFING FRAMES BETWEEN CONTAINERS ARE NOT REQUIRED.
   b. WHEN A STACK OF MOD 1 CONTAINERS IS THREE WIDE IT MAY NOT BE POSSIBLE TO NAIL THE SLEEPERS (PIECE 10, PAGE 3 OR PIECE 7, PAGE 5) TO THE TRAILER FLOOR BECAUSE OF THE TRAILER DESIGN. WHEN THIS HAPPENS SUBSTITUTE THE METHOD SHOWN BELOW FOR EACH SLEEPER THAT CANNOT BE NAILED TO THE TRAILER FLOOR.
2. STUB STAKE IS SIZED TO FIT THE STAKE POCKET AND CUT TO EXTEND APPROXIMATELY 9 INCHES ABOVE TRAILER FLOOR AND 3 INCHES BELOW STAKE POCKET. DRIVE ONE 20d NAIL BELOW STAKE POCKET ALLOWING ONE INCH TO PROTRUDE TO PREVENT STAKE WORKING OUT OF PLACE.
3. THICKNESS OF FILLER SHALL BE SUFFICIENT TO FILL SPACE BETWEEN STAKE AND CONTAINER SKIDS. IT MAY BE COMPOSED OF ONE OR MORE PIECES. WHEN TWO OR MORE PIECES ARE USED, THEY SHALL BE LAMINATED TOGETHER.





DETAIL J

STUB STAKE AND FILLER ASSEMBLY
SIDE BLOCKING METHOD

REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140 - N681)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.